

Compiler Design

Class test

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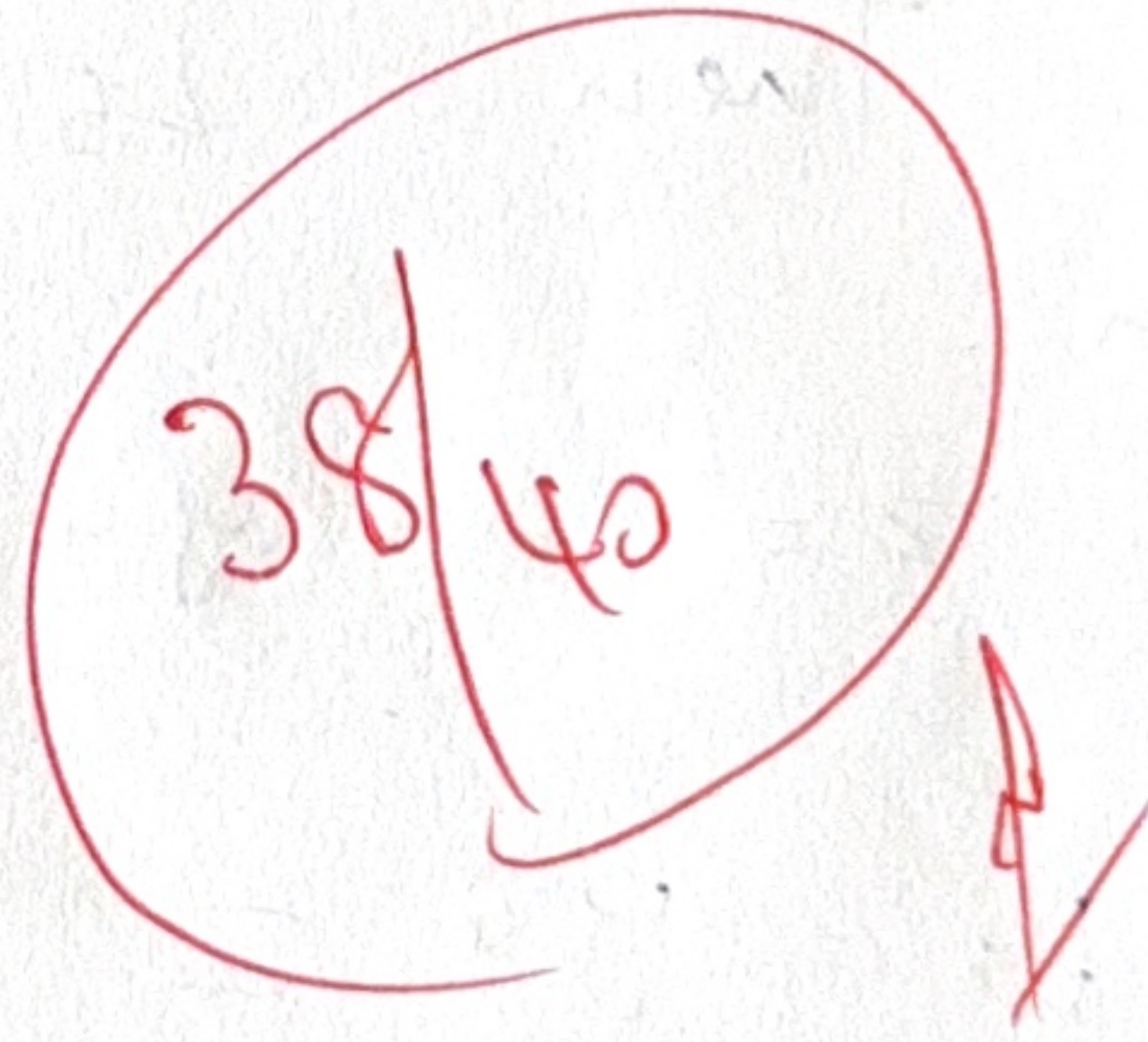
- ① Step 1: Condition identification $(a < b)$ and $(c < d)$ or $(a > d)$ $(a < b)$ and $((c < d) \text{ or } (a > d))$

Condition

$a < b$

$c < d$

$a > d$



Step 2: Table Creation

L1 L2 L3 L4 L5 L6

Step 3: TAC for Condition

L1;

if $a < b$ goto L2

goto L1

L2

if $c < d$ goto L4

goto L3

L3

if $a > d$ goto L4

goto L5

if $a < b$ is false, the complete and expression is false, so control goes to the else part. If $a < b$ is true expression is evaluated. If either $c < d$ or $a > b$ is true.

Step 4: Generate TAC for Then Part

The statement is $Z = x + y + z$.
Multiplication has higher precedence.

Then addition

L4 Then L5 else

Then :

$$Z = x + y + z$$

L4

$$t_1 = y * z$$

$$t_2 = x - t_1$$

$$Z = t_2$$

goto L6

Else :

$$Z = z + 1$$

L5

$$t_3 = z + 1$$

$$Z = t_3$$

L6 :

Final TAC :

L1 :

if $a < b$ goto L2

goto L5

L2

if $c < d$ goto L4

goto L2

L3

if $a > b$ goto L_4

goto L_5

L_5

$t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

goto L_6 :

② Given boolean expression

$(a < b)$ and $(c < d)$ or $(a < d)$

Step 1 : Apply operator precedence

AND has higher precedence

than OR, so the expression is

interpreted as :

$[(a < b) \text{ and } (c < d)] \text{ or } (a < d)$

Step 2 : Generate intermediate jump

if $a < b$ goto -

goto -

if $c < d$ goto -

goto -

if $c < d$ goto -

goto -

Step 3 : HANDLE AND

$(a < b) \text{ and } (c < d)$ true if $a < b$ is true control must continue to second condition.

BackPath Statement 1 to Statement 3.

1. if $a < b$ goto 5.

Step 4: Handle OR

If the first and expression is false the second or operand $e < f$ must be evaluated.

Back Path Statement 2 to statement 5

Back Path Statement 4 to statement 5.

Step 5: Final Back Code

1. if $a < b$ goto 3

2. goto 5.

3. if $c < d$ goto 7

4. goto 5

5. if $e < f$ goto 7

6. goto 8

7. True

8. False

③ $[(a > b) \text{ OR } (c < d)] \text{ OR } (e < f)$

Because This is an OR expression. if any one of the three relation condition is true, the compare expression is true. check the first condition

L1:

if $a > b$ goto True.

goto L2

check the second condition

L₁:

if $c < d$ goto TRUE

goto L₃

check the third condition

L₃

if $e < f$ goto L TRUE

goto L FALSE

Final TAC for a

L₁

if $a > b$ goto TRUE

goto L₂

goto L₃

L₃

if $c < f$ goto L TRUE

goto FALSE

L TRUE

goto AND

L END

While $(A < C \text{ and } B > 0)$ do

if $A = 1$ then $C = C + 1$

else while $A \leq B$ do

$A = A + B$

L₁

L₁ = C + 1

C = t₁

goto L₁

L₂

t₂ = A + B

A = t₂

L₃

t₃ = A - B

if A == 1 goto L₄

goto L₅

L₄

t₁ = C + 1

C = t₁

goto L₁

L₅

if A < D goto L₆

goto L₅

L₆

t = A + B

A = t₂

goto L₅

L END